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Next item →

1. Consider random variable X having Gaussian distribution with mean of 1 and variance of 2, random variable Y having Gaussian distribution with mean of 2 and variance of 3. That is, we can say that $X \sim \mathcal{N}(1, 2)$ and $Y \sim \mathcal{N}(2, 3)$. Also consider random variable $Z = 3X + 2Y$.

1 / 1 point

What is $\mathbb{E}[Z]$?

Hint: All numeric answers in this quiz are integers.

✔ Correct

2. Using X , Y , and Z from problem 1, if X and Y are uncorrelated, what is the variance of Z ? That is, what is $\Sigma_{\tilde{Z}}$?

1 / 1 point

✔ Correct

3. Again using X , Y , and Z from problem 1, if X and Y are independent, what is $\mathbb{E}[Z \mid X = 2]$?

1 / 1 point

✔ Correct

4. What was the critical assumption that we made when finding an equation to express $\mathbb{E}[\tilde{x}_k^- \mid z_k]$ as part of the generic Gaussian sequential probabilistic inference solution?

1 / 1 point

✔ Correct

5. What are the uses of the Kalman-filter gain factor L_k ? (Select all that apply.)

1 / 1 point

✔ Correct

6. Which of the following assumptions did we make when deriving the linear Kalman filter steps? Select all that apply.

1 / 1 point

✔ Correct

7. To compute $\hat{x}_k^-$, $\Sigma_{\tilde{x},k}^-$, and $\hat{z}_k$, what information do we require? Select all that apply.

1 / 1 point

✔ Correct

8. To compute $\hat{x}_k^+$ and $\Sigma_{\tilde{x},k}^+$, what information do we require? Select all that apply.

1 / 1 point

✔ Correct

9. What are possible reasons for a large entry in the Kalman gain matrix L_k ? Select all that apply.

1 / 1 point

✔ Correct

10. What is the significance of the state estimation error covariance matrix $\Sigma_{\tilde{x},k}^+$?

1 / 1 point

✔ Correct